

# Prateek Anand

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My research interests are in developing novel AI/ML computational methods. Broadly, I am invested in deep generative models, traditional machine learning, and statistical approaches for understanding and generating biological data.

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| UCLA   <i>Ph.D., Computer Science</i>   Advisor: Dr. Sriram Sankararaman                                                                                 | 2024 - 2029 |
| • Latest work, GPC (deep generative model for genetic data): <i>PLOS Genetics</i> ; <b>Top 10% Oral Spotlight</b> , <i>ICLR Gen<sup>2</sup> Workshop</i> |             |
| UCLA   <i>M.S., Computer Science</i>                                                                                                                     | 2024 - 2026 |
| UCLA   <i>B.S., Computer Science</i>                                                                                                                     | 2020 - 2024 |

## PROFESSIONAL EXPERIENCE

|                                                                                                                                 |                     |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------|
| UCLA Computer Science   Sriram Lab                                                                                              | Jun 2022 - Present  |
| Graduate Student Researcher                                                                                                     |                     |
| • Developing discrete diffusion models and improved guidance methods for cell-type-specific regulatory DNA design               |                     |
| • Building deep generative models of genetic variation for synthetic data and imputation                                        |                     |
| • Creating scalable machine learning/statistical software to understand human genetic architecture                              |                     |
| Research Intern (Bruins in Genomics Research Program)                                                                           |                     |
| • Evaluated a symbolic regression-based explanation method for complex black box machine learning models                        |                     |
| Stanford University School of Medicine   Curtis Lab                                                                             | Jun 2023 - Jan 2024 |
| Research Intern (Canary CREST Research Program for Early Cancer Detection)                                                      |                     |
| • Computational modeling and inference, refined model built using approximate Bayesian computation to detect blood cancer early |                     |
| UCLA Jonsson Comprehensive Cancer Center   Boutros Lab                                                                          | Jan 2021 - Apr 2022 |
| Research Assistant                                                                                                              |                     |
| • Updated, maintained, and evaluated CI/CD HATCHet pipeline for use on lab cluster compute server                               |                     |

## PUBLICATIONS

\* denotes equal contribution

- **Prateek Anand**, Anji Liu, Meihua Dang, Boyang Fu, Xinzhu Wei, Guy Van den Broeck\*, Sriram Sankararaman\*. GPC: An expressive and tractable deep generative model for genetic variation data, *PLOS Genetics*, (2026).
- Yash Patel\*, Chenghao Zhu\*, Takafumi N. Yamaguchi\*, ..., **Prateek Anand**, ..., Paul C. Boutros. Metapipeline-DNA: A comprehensive germline and somatic genomics Nextflow pipeline, *Cell Reports Methods*, (2026).
- Boyang Fu\*, Ali Pazokitoroudi\*, ..., **Prateek Anand**, ..., Sriram Sankararaman. A biobank-scale test of marginal epistasis reveals genome-wide signals of polygenic interaction effects, *Nature Genetics*, (2025).
- Boyang Fu\*, **Prateek Anand**\*, Aakarsh Anand\*, Joel Mefford, Sriram Sankararaman. A scalable adaptive quadratic kernel method for interpretable epistasis analysis in complex traits, *Genome Research*, (2024). Also presented at *RECOMB* (2024).

## PREPRINTS

- **Prateek Anand**, ..., Guy Van den Broeck\*, Sriram Sankararaman\*. GPC: Deep generative model of genetic variation data improves imputation accuracy in private populations, *ICLR Gen<sup>2</sup> Workshop*, (2026). **Top 10% Oral Spotlight Presentation.**
- Zhengtong Liu, Boyang Fu, Moonseong Jeong, **Prateek Anand**, ..., Sriram Sankararaman. Comprehensive gene heritability estimation reveals the genetic architecture of rare coding variants underlying complex traits, *bioRxiv*, (2025).

## PRESENTATIONS

- **Prateek Anand**, ..., Guy Van den Broeck\*, Sriram Sankararaman\*. Deep generative model of genetic variation data improves imputation accuracy in private populations, *American Society of Human Genetics*, (2025).

## TEACHING

- **Machine Learning Applications in Genetics (UCLA CS 124)**: Teaching Assistant (Spring 2026)
- **Machine Learning Applications in Biomedicine (UCLA DSB 205)**: Teaching Assistant (Winter 2026)
- **Introduction to Machine Learning (UCLA CS 146)**: Teaching Assistant (Winter 2025); Reader (Winter 2023, Winter 2024)

## SKILLS

- **Programming & Tools**: Python, C++, PyTorch, Scikit-learn, NumPy, Pandas, Git, Cluster Computing
- **Expertise**: Deep Learning, Generative Models, Diffusion Models, Machine Learning, Statistics, Probabilistic Models

## AWARDS

- **Google PhD Fellowship Nominee**: One of four nominated by UCLA (2026)
- **Warren Alpert Computational Biology and AI Fellow**: Graduate Training Fellowship (2024)
- **NCI Scholarship**: Canary CREST Program (2023); **NSF REU Scholarship**: Bruins in Genomics Program (2022)
- **Andy Grove Intel Scholarship**: Academic Excellence (2020); **UCLA Dean's Honors List**: 6 quarters (2020 - 2024)